Lab Task 3: Writing a Unique Essay Using LaTeX Commands

Rakibul Islam Rafi
Student ID: 231-115-109
Section: C
Course Code: CSE 429

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Abstract

This report presents a detailed discussion on Artificial Intelligence (AI) and its important subfields. It explains mathematical foundations, neural network structures, and computational requirements of modern intelligent systems. Various LaTeX commands are used throughout the report to ensure professional formatting and mathematical clarity.

1 Introduction

Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence. A learning model can be represented mathematically as:

$$f_\theta : X \to Y, \quad \theta \in \mathbb{R}^d \tag{1}$$

where X is the input space, Y is the output space, and θ represents the learnable parameters.

2 Structural Foundation: Neural Networks

Neural networks consist of interconnected layers:

- **Input Layer** – Receives raw data $x \in \mathbb{R}^n$
- **Hidden Layers** – Apply transformation:

$$h^{(\ell)} = \sigma\left(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}\right) \tag{2}$$

- **Output Layer** – Produces final prediction $\hat{y}$

2.1 Backpropagation and Gradient Descent

Model parameters are updated using Gradient Descent:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta_t) \tag{3}$$

where η is the learning rate and ∇ represents the gradient.

3 Specialised Architectures and Applications

Model	Application	Example
CNN	Image Processing	Facial Recognition
RNN	Sequential Data	Speech Recognition
Transformer	NLP	Language Translation
GAN	Data Generation	Image Synthesis

Table 1: Major AI Architectures

3.1 Convolutional Neural Networks (CNNs)

$$(I * K)(i, j) = \sum_m \sum_n I(i + m, j + n)K(m, n) \tag{4}$$

3.2 Recurrent Neural Networks (RNNs)

$$h_t = \tanh(W_h h_{t-1} + W_x x_t) \tag{5}$$

4 Computational Requirements and Scalability

The relationship between data size and generalization error is approximately:

$$\epsilon(N) \approx C \cdot N^{-\alpha}, \quad \alpha > 0 \tag{6}$$

Key hardware used:

- GPUs
- TPUs

5 Mathematical Symbol Reference

- ∂ – Partial derivative
- ∇ – Gradient
- $\sum$ – Summation
- $\int$ – Integral
- $\forall$ – For all
- $\exists$ – There exists
- $\in$ – Element of
- $\alpha, \beta, \theta, \sigma$ – Greek symbols

6 Conclusion

Artificial Intelligence is transforming modern technology. With mathematical modeling, neural architectures, and scalable computation, AI systems are becoming more powerful and efficient. This lab task demonstrates how LaTeX can be used to create a structured and professional academic report.

7 References

1. LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep Learning. Nature.
2. Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.

Description of Commands Used

In this lab task, several LaTeX commands were used such as , , , , ,

8 ,

8.1 ,

math mode using ... and equation environment, table environment, itemize environment, and

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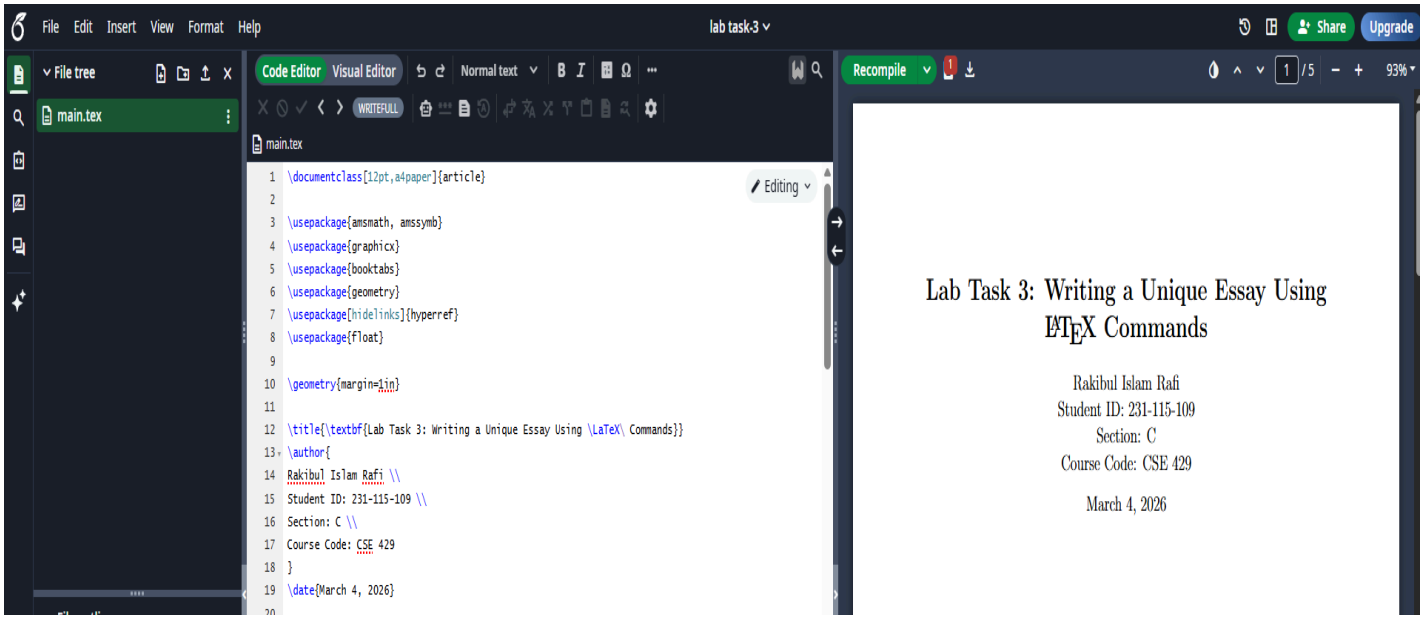
to create a structured and professional academic report.

Other Features Used

Automatic page numbering, proper equation alignment, professional table formatting, structured headings, and reference formatting were used to enhance document quality.

Challenges Faced

During this task, challenges included writing complex equations correctly, aligning tables properly, managing mathematical symbols, and maintaining professional formatting. These challenges were overcome through careful revision and practice.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\documentclass[12pt,a4paper]{article}
\usepackage{amsmath, amssymb}
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\usepackage{booktabs}
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\author{
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Course Code: CSE 429
}
\date{March 4, 2026}
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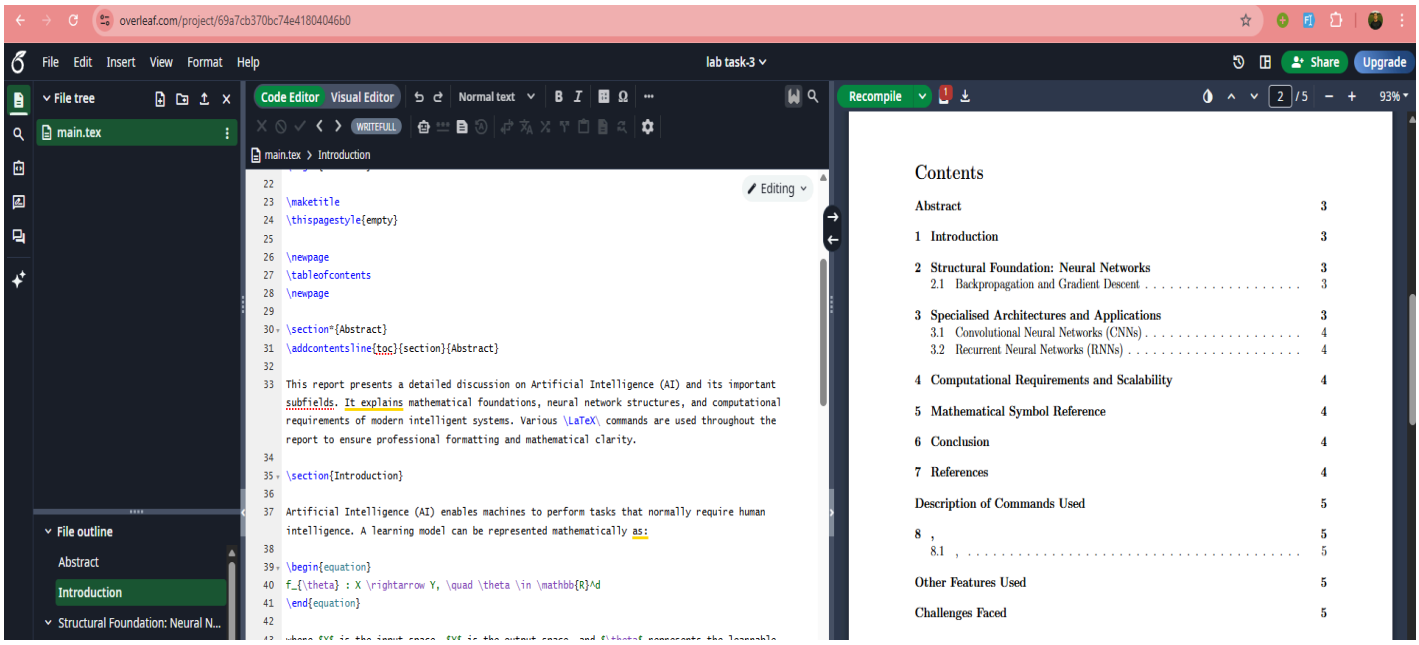
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\section*{Abstract}
\addcontentsline{toc}{section}{Abstract}
This report presents a detailed discussion on Artificial Intelligence (AI) and its important subfields. It explains mathematical foundations, neural network structures, and computational requirements of modern intelligent systems. Various \LaTeX\ commands are used throughout the report to ensure professional formatting and mathematical clarity.
\section{Introduction}
Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence. A learning model can be represented mathematically as:
\begin{equation}
f_{\theta} : X \rightarrow Y, \quad \theta \in \mathbb{R}^d
\end{equation}
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3 Specialised Architectures and Applications 3
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4 Computational Requirements and Scalability 4
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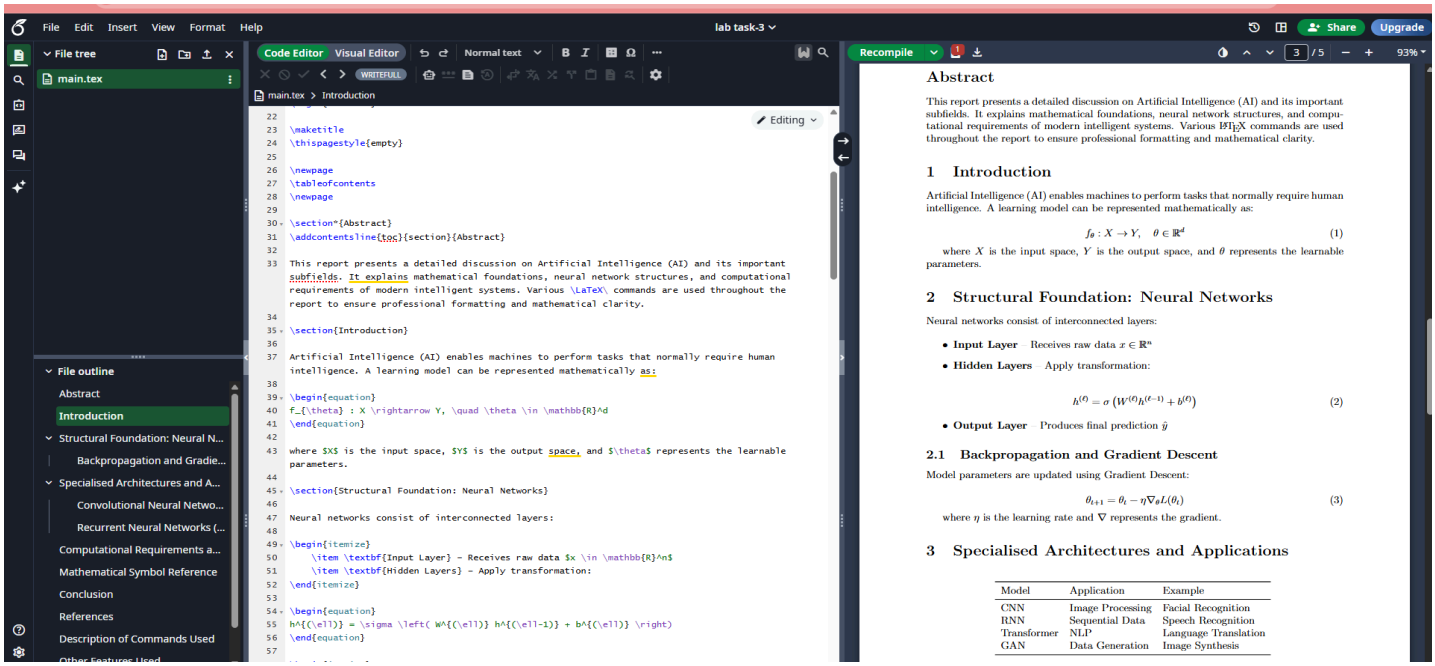
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where X is the input space, Y is the output space, and θ represents the learnable parameters.
\section{Structural Foundation: Neural Networks}
Neural networks consist of interconnected layers:
\begin{itemize}
\item \textbf{Input Layer} – Receives raw data $x \in \mathbb{R}^n$
\item \textbf{Hidden Layers} – Apply transformation:
\end{itemize}
\begin{equation}
h^{(\ell)} = \sigma \left(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)} \right)
\end{equation}
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Computational Requirements a...
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2 Structural Foundation: Neural Networks
Neural networks consist of interconnected layers:
Input Layer – Receives raw data $x \in \mathbb{R}^n$
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$h^{(\ell)} = \sigma \left(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)} \right)$ (2)
Output Layer – Produces final prediction $\hat{y}$
2.1 Backpropagation and Gradient Descent
Model parameters are updated using Gradient Descent:
$\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta_t)$ (3)
where η is the learning rate and ∇ represents the gradient.
3 Specialised Architectures and Applications
Model | Application | Example
CNN | Image Processing | Facial Recognition
RNN | Sequential Data | Speech Recognition
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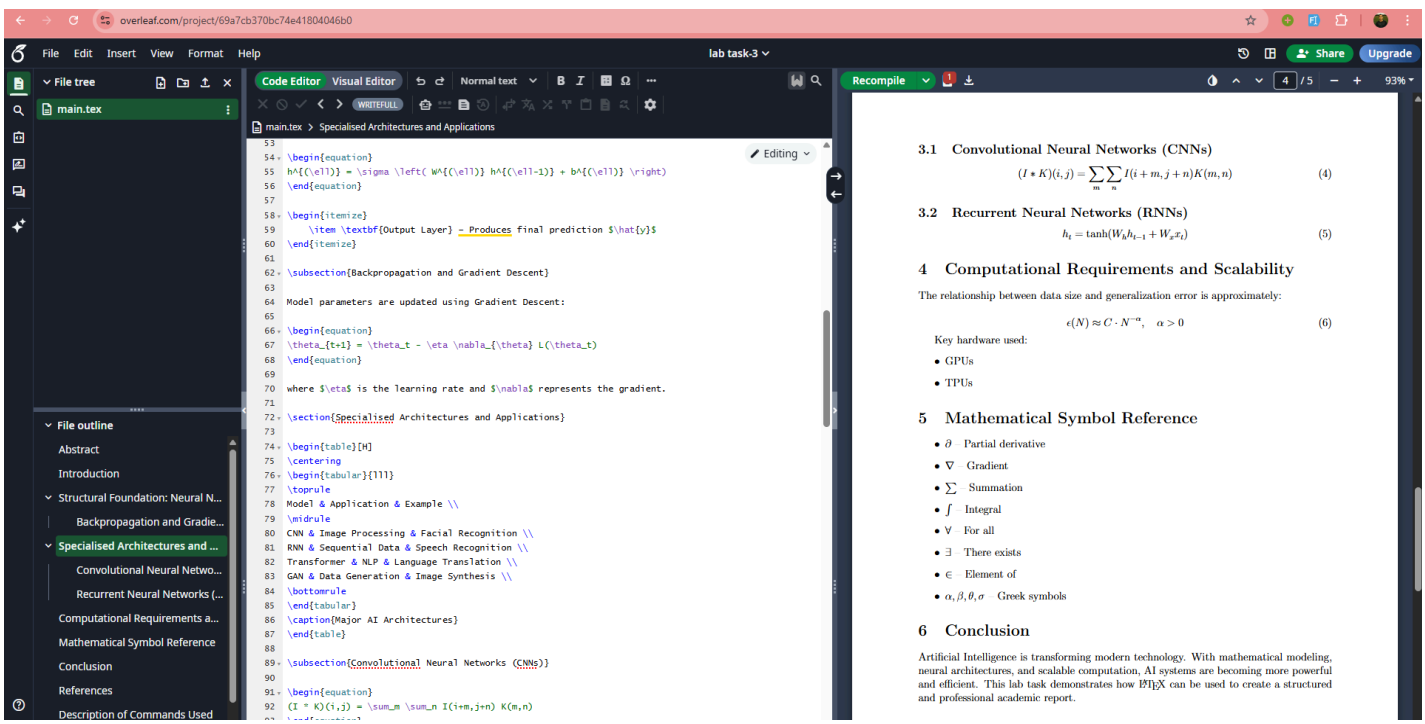
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\begin{equation}
h^{(\ell)} = \sigma \left(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)} \right)
\end{equation}
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\end{itemize}
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Model parameters are updated using Gradient Descent:
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\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t)
\end{equation}
where η is the learning rate and ∇ represents the gradient.
\section{Specialised Architectures and Applications}
\begin{table}[H]
\centering
\begin{tabular}{lll}
\toprule
Model & Application & Example \\
\midrule
CNN & Image Processing & Facial Recognition \\
RNN & Sequential Data & Speech Recognition \\
Transformer & NLP & Language Translation \\
GAN & Data Generation & Image Synthesis \\
\bottomrule
\end{tabular}
\caption{Major AI Architectures}
\end{table}
\subsection{Convolutional Neural Networks (CNNs)}
\begin{equation}
(I * K)(i,j) = \sum_m \sum_n I(i+m, j+n) K(m,n)
\end{equation}
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3.1 Convolutional Neural Networks (CNNs)
$(I * K)(i,j) = \sum_m \sum_n I(i+m, j+n) K(m,n)$ (4)
3.2 Recurrent Neural Networks (RNNs)
$h_t = \tanh(W_h h_{t-1} + W_x x_t)$ (5)
4 Computational Requirements and Scalability
The relationship between data size and generalization error is approximately:
$\epsilon(N) \approx C \cdot N^{-\alpha}, \quad \alpha > 0$ (6)
Key hardware used:
GPUs
TPUs
5 Mathematical Symbol Reference
∂ – Partial derivative
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$\sum$ – Summation
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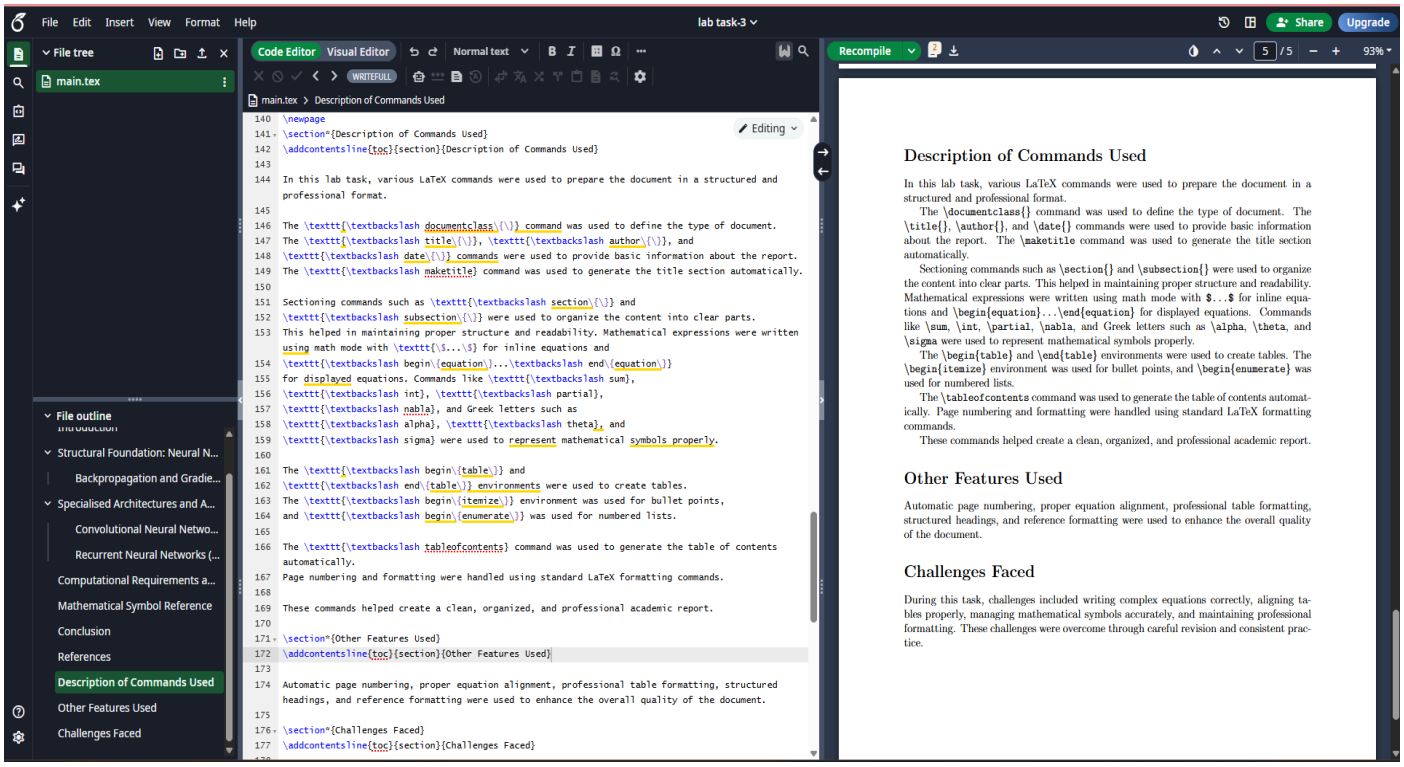
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\section*{Description of Commands Used}
\addcontentsline{toc}{section}{Description of Commands Used}
In this lab task, various LaTeX commands were used to prepare the document in a structured and professional format.
The \texttt{\textbackslash documentclass\{\}} command was used to define the type of document.
The \texttt{\textbackslash title\{\}}, \texttt{\textbackslash author\{\}}, and
\texttt{\textbackslash date\{\}} commands were used to provide basic information about the report.
The \texttt{\textbackslash maketitle} command was used to generate the title section automatically.
Sectioning commands such as \texttt{\textbackslash section\{\}} and
\texttt{\textbackslash subsection\{\}} were used to organize the content into clear parts.
This helped in maintaining proper structure and readability. Mathematical expressions were written using math mode with \texttt{\$...\$} for inline equations and
\texttt{\textbackslash begin\{equation\}...\textbackslash end\{equation\}}
for displayed equations. Commands like \texttt{\textbackslash sum},
\texttt{\textbackslash int}, \texttt{\textbackslash partial},
\texttt{\textbackslash nabla}, and Greek letters such as
\texttt{\textbackslash alpha}, \texttt{\textbackslash theta}, and
\texttt{\textbackslash sigma} were used to represent mathematical symbols properly.
The \texttt{\textbackslash begin\{table\}} and
\texttt{\textbackslash end\{table\}} environments were used to create tables.
The \texttt{\textbackslash begin\{itemize\}} environment was used for bullet points,
and \texttt{\textbackslash begin\{enumerate\}} was used for numbered lists.
The \texttt{\textbackslash tableofcontents} command was used to generate the table of contents automatically.
Page numbering and formatting were handled using standard LaTeX formatting commands.
These commands helped create a clean, organized, and professional academic report.
\section*{Other Features Used}
\addcontentsline{toc}{section}{Other Features Used}
Automatic page numbering, proper equation alignment, professional table formatting, structured headings, and reference formatting were used to enhance the overall quality of the document.
\section*{Challenges Faced}
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The \begin{table} and \end{table} environments were used to create tables. The \begin{itemize} environment was used for bullet points, and \begin{enumerate} was used for numbered lists.
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Challenges Faced
During this task, challenges included writing complex equations correctly, aligning tables properly, managing mathematical symbols accurately, and maintaining professional formatting. These challenges were overcome through careful revision and consistent practice.